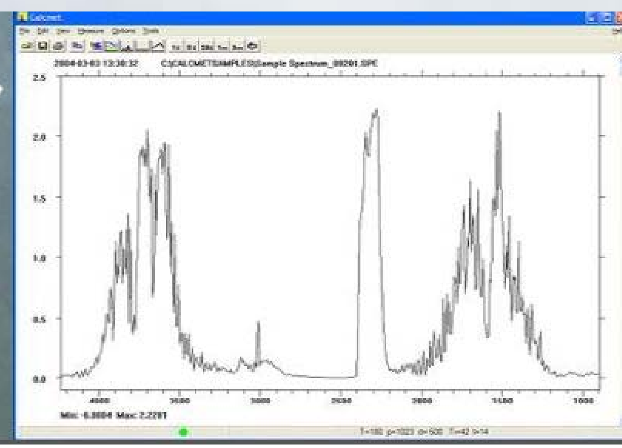
FTIR Gas Analysis

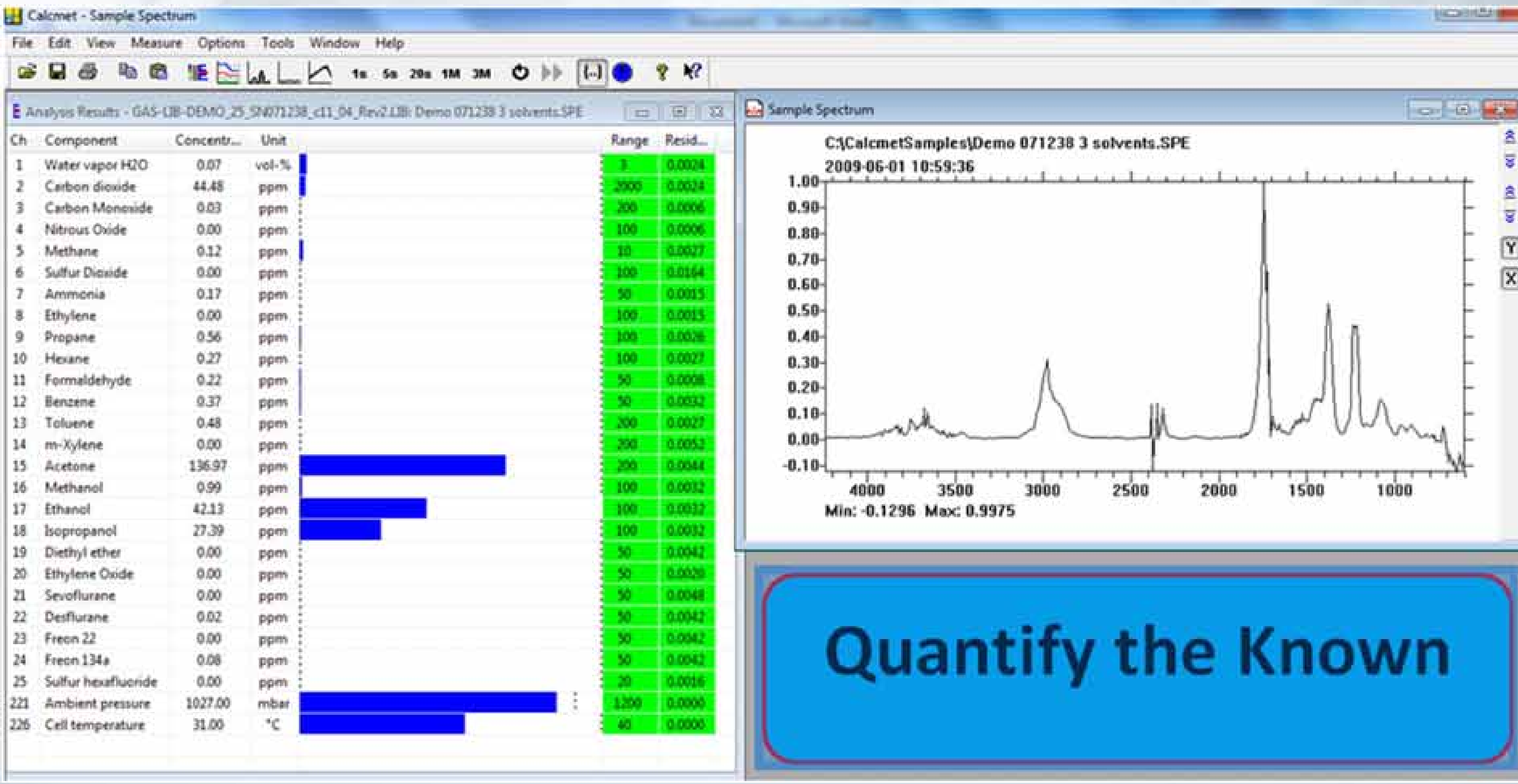
Joel Myerson
Joel@ESafetyInc.com



Gas phase infrared spectroscopy

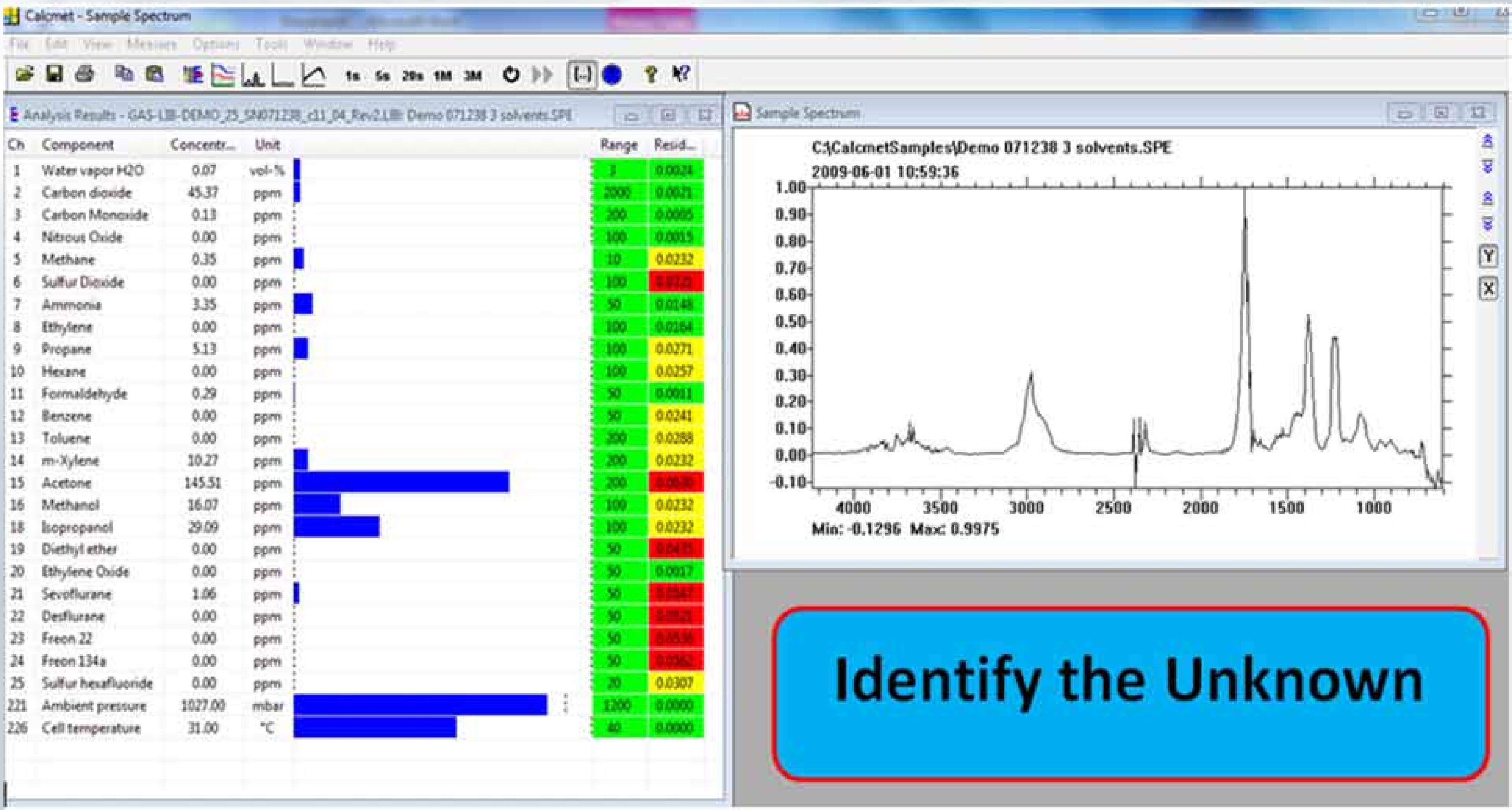
- Molecules in gas phase vibrate and rotate at frequencies characteristic to each molecule. Each frequency is associated with an energy state of a molecule
- Infrared radiation moves the molecules to higher energy states; characteristic frequencies are absorbed by the molecule in the process
- Each molecule absorbs infrared radiation at several characteristic frequencies (wavelengths)
- The result is an IR absorption spectrum; a fingerprint unique to each molecule

A. Dual Functionality of FTIR

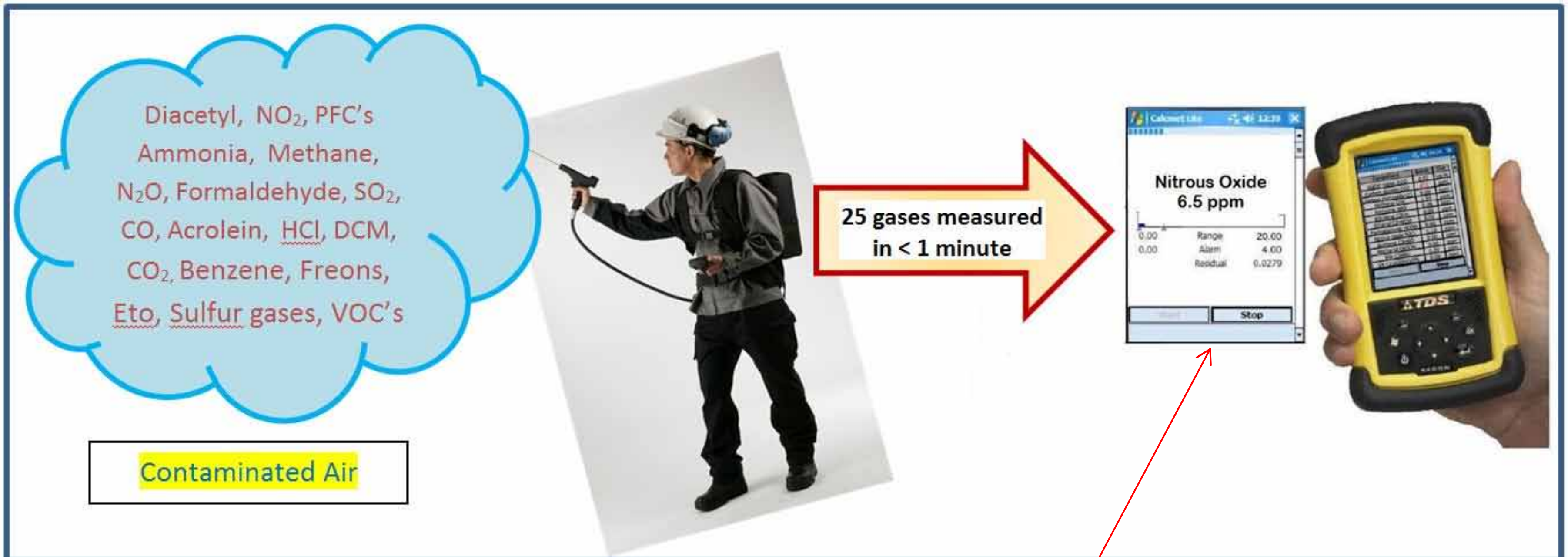


Gasmet

B. Dual Functionality of FTIR



Understanding the power of FTIR[1] gas analysis



FTIR Gas Analysis using a combination of optical light measurement and a mathematical algorithm to measure many gases and their concentrations simultaneously.

The **25** gases can be chosen & changed from a master library (**237** gases).

1. FTIR = Fourier Transform Infrared

Primary Site Safety Assessment

- **FTIR Will never replace Gas Detector**
- **(<u>Oxygen, Hydrogen Sulfide, CO & LEL</u>)**

What next if no alarms on the Gas Detector ?
Is the site safe from Toxic Gas ?
Is the Gas Detector giving a false sense of security ?

Many gases are toxic at levels below the range of a gas detector .

FTIR Gas Analyzer can rapidly assess site safety & exposure risk on daily basis to toxics.

Applications include :

- Detecting **Natural Gas leaks** at low levels (Methane (ppm) & Mercaptans)
- Detecting **Ammonia leaks**
- Toxics at Clandestine Lab. clean up site
- Measuring for the presence of **Freons** at industrial sites
- **Acid gases** such as **HCl, HF** and **HCN, Nitric Acid**
- **Screening for TIC's, TIM's & CWA's**
- **Chemicals** – Methyl Bromide, Chloropicrin, Vikane, Ethylene Oxide
- **Refinery Toxics** – Benzene, Styrene, Aldehydes, Hexane, Methanol, Carbon Disulfide,
- **Semiconductor Plants** – Arsine, Phosphine, Silanes, Boron
- **Hospitals** – Formaldehyde, Hydrogen Peroxide, Ethylene Oxide, Peracetic Acid
- **IAQ investigations**
- **LEED certification**

Quantify the "Known"

Applications include

1. Detecting **Natural Gas leaks** at low levels
2. Detecting **Ammonia leaks**
3. Assessing build up of **Carbon Monoxide**
4. Measuring for the presence of **Freons** at industrial sites
5. **Acid gases** such as **HCl, HF** and **HCN**
6. **Screening for TIC's, TIM's & CWA's**
7. Inorganics including Ammonia, Phosphine, Oxides of Nitrogen, SO_2
8. **Declaring All Clear at incident site**

Clandestine lab. Measurements

Gases	Range (ppm)
Ammonia	0 – 100
Methylamine	0 – 200
Chloroform	0 – 100
Phosphine	0 – 50
Benzene	0 – 50
Toluene	0 – 200

Sample Lower Detection Limits – 60 seconds

Chemical	Det. range	LDL (ppm)
• **Acetaldehyde** C_2H_4O	0 - 500	0.13
• **Acetic Acid** CH_3COOH	0 – 100	0.04
• **Acetone** CH_3COCH_3	0 - 1000	0.07
• **Acrolein** C_3H_4O	0 - 5	0.25
• **Acrylonitrile** C_3H_3N	0 - 200	0.35
• **Ammonia** NH_3	0 - 100	0.13
• **Aniline** C_6H_7N	0 -50	0.06
• **Arsine** AsH_3	0 - 50	0.02
• **Benzene** C_6H_6	0 - 20	0.13
• **Boron Trichloride** BCl_3	0 -50	0.01

LDLs for similar chemicals

Chemical	Det. Range	LDL (ppm)
Freon 113 (CFC-113) $C_2F_3Cl_3$	0 - 2000	0.02
Freon 114 (CFC-114) $C_2F_4Cl_2$	0 - 2000	0.01
Freon 12 (CFC-12) CCl_2F_2	0 - 2000	0.02
Freon 134a (HFC-134A) $C_2H_2F_4$	0 - 2000	0.01
Freon 141b $C_2H_3FCl_2$	0 - 2000	0.07
Freon 22 (HCFC-22) $CHClF_2$	0 - 2000	0.01

Test the Air for toxic gases before entering... building, house, container,........

- Fumigants

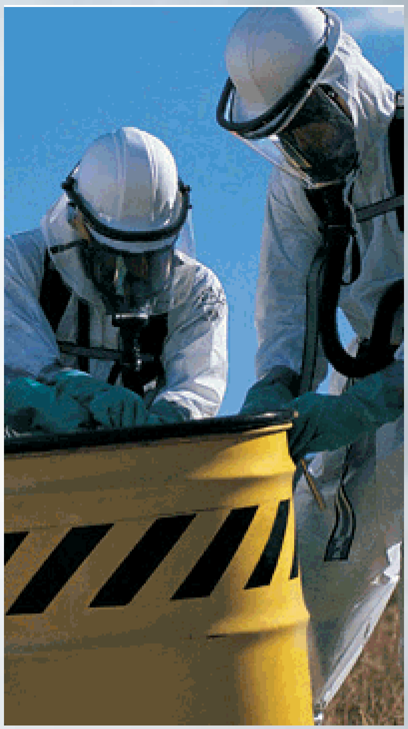
Methyl Bromide
Vikane
Phosphine
Hydrogen Cyanide

Gasmet

Testing for Toxic Industrial Chemicals

Monitor the following 25 gases simultaneously -

1. Acrolein – **(0.25 ppm)**
2. Acrylonitrile - (**0.35 ppm)**
3. Ammonia - **(0.13 ppm)**
4. Arsine - **(0.02 ppm)**
5. Benzene - **(0.12 ppm)**
6. Boron trichloride – **(0.01 ppm)**
7. Carbon dioxide – **(< 10 ppm)**
8. Carbon monoxide – **(0.25 ppm)**
9. Carbon Disulfide - **(0.17 ppm)**
10. Dichloromethane – **(0.13 ppm)**
11. Ethylene oxide - **(0.17 ppm)**
12. Formaldehyde – **(0.09 ppm**)
13. Hydrogen chloride – **(0.20 ppm)**
14. Hydrogen cyanide – **(0.35 ppm)**
15. Hydrogen fluoride – **(0.30 ppm)**
16. Methane - **(0.06 ppm)**
17. Methyl Mercaptan – **(0.42 ppm)**
18. Nitrogen dioxide – **(0.37 ppm)**
19. Nitrous oxide – **(0.02 ppm)**
20. Phosgene – **(0.02 ppm**)
21. Phosphine – (**0.20 ppm)**
22. Sulfur dioxide – **(0.03 ppm)**
23. Sulfuryl fluoride – **(0.03 ppm)**
24. Toluene – **(0.13 ppm)**
25. Water Vapour



A. Dual Functionality of FTIR

Protecting health

Or select from a calibration library to create site specific gas calibration libraries

Testing for CWA's

#	Compound name	Formula	CAS number	Standard range	
	DX4030 Chemical Warfare Agents	GAS-LIB-005			
	Standard components				
1	Water	H_2O	7732-18-5	3	vol-%
2	Carbon dioxide	CO_2	124-38-9	2000	ppm
3	Carbon monoxide	CO	630-08-0	50	ppm
4	Nitrous oxide	N_2O	10024-97-2	50	ppm
5	Sulfur dioxide	SO_2	7446-09-5	50	ppm
6	Ammonia	NH_3	7664-41-7	50	ppm
7	Ethylene (Ethene)	C_2H_4	74-85-1	50	ppm
8	Formaldehyde	$HCOH$	50-00-0	50	ppm
9	Acetone	CH_3COCH_3	67-64-1	50	ppm
10	Methanol	CH_3OH	67-56-1	50	ppm
11	Ethanol	C_2H_5OH	64-17-5	50	ppm
12	Isopropanol (2-Propanol; Isopropyl alcohol)	$CH_3CHOHCH_3$	67-63-0	50	ppm
13	Methane	CH_4	74-82-8	50	ppm
14	Phosgene	$COCl_2$	75-44-5	50	ppm
15	Hydrogen cyanide	HCN	74-90-8	50	ppm
16	VX (Methylphosphonothioic acid)	$C_{11}H_{26}NO_2PS$	50782-69-9	10	ppm
17	Lewisite (2-Chlorovinyldichloroarsine)	$C_2H_2AsCl_3$	541-25-3	20	ppm
18	Chlorosoman (1,2,2-Trimethyl propyl methyl phosphonochloridate)	$C_7H_{16}O_2PCl$	7040–57–5	20	ppm
19	Sarin (*o*-Isopropyl methylphosphonofluoridate; GB)	$(CH_3)_2CHOP(CH_3)FO$	107-44-8	10	ppm
20	Soman (*o*-Pinacolyl methylphosphonofluoridate; GD)	$C_7H_{16}FO_2P$	96-64-0	10	ppm
21	Mustard gas (Bis(2-chloroethyl)sulphide; HD)	$(C_2H_4Cl)_2S$	505-60-2	20	ppm
22	Tabun (*o*-Ethyl N,N-dimethyl phosphoramidocyanidate; GA)	$C_5H_{11}N_2O_2P$	77-81-6	10	ppm
23	Chloropicrin (PS)	CCl_3NO_2	76-06-2	20	ppm
24	Dimethyl phosphite (Dimethyl hydrogen phosphite)	$C_2H_7O_3P$	868-85-9	10	ppm
25	Dimethyl methylphosphonate (DMMP)	$C_3H_9O_3P$	756-79-6	10	ppm

C B R N E

Gasmet

Testing for **Post-Fire Toxic Gases**

"Removal of respiratory protection during fire overhaul activities or in the general area can expose firefighters and fire investigators to an **unknown variety of toxic chemicals** and particulates."

Chemical Exposures	OSHA TWA PEL (ppm)	# Fires Analyzed	# fires found
Acrolein	0.1	11	4
Aldehydes (total aliphatic)	n.d.	29	19
Ammonia	50	29	8
Benzene	1	29	10
Benzyl chloride	1	12	3
Carbon disulfide	30	3	3
Carbon monoxide	50	38	30
Formaldehyde	0.75	29	4
Furfural	5	3	3
Glutaraldehyde	n.d.	12	12
Hydrogen chloride	5	37	8
Naphthalene	10	37	7
Nitrogen dioxide	3	37	28
Nitrogen monoxide	25	29	28
Ozone	0.1	29	21
Phenol	5	29	9
Sulfur dioxide	5	29	2
Styrene	100	29	25
Toluene	200	29	27

A portable FTIR Gas Analyzer can quickly test for these gases and alert firefighter if site is clear to remove SCBA.

Paper by_Deric C. Weiss and Jeff T. Miller Tualatin Valley Fire & Rescue 2011

Gasmet

Identify the Unknown

Case Study

MI
Region2
Hazmat
Team

On November 30, 2012, Region 2 South TSRT Coordinator, was notified of a potential gas or vapor leak in the Human Pathology Lab at St. Joseph Hospital. Having just received and programmed our new laptop computer for the **DX-4040 FTIR**, we decided to take advantage of the opportunity for a training deployment of the TSRT and the Gasmet.

We were advised that the **suspicion was increased levels of Formaldehyde.** We ran tests in five different work areas within the lab. With the FTIR gas analyzer we were able to simultaneously test for 50 different gases/vapors using quantitative and qualitative analysis. **The results of our tests revealed that the substance was Toluene and not Formaldehyde.** The levels were well within safe limits of all found substances as well as TWA's [Time Weighted Averages] **The FTIR Gas Analyzer performed flawlessly and proved it's capability beyond doubt.** The TSRT had a successful training deployment and had many educational benefits from the experience.

Case Study

HAZMAT team at Owosso hospital after man ingests substance

by NBC25 Newsroom

Posted: 09.13.2013 at 12:56 PM

Updated: 09.13.2013 at 1:30 PM

A patient suspected of ingesting potassium cyanide which has the potential for an off-gassing of **HCN** arrived at Owosso Hospital Emergency Department.

A hazardous material tent was set up outside of the emergency room near the ambulance entrance for staff treating the patient.

The patient’s breathe was monitored with a FTIR Gas Analyzer for presence of **HCN** and other toxic gases.

“Our primary reason for diverting ambulance traffic was to allow our staff, and those agency personnel, to focus on ensuring that a safe environment was maintained.” Hospital CEO

Identify the Unknown

Case Study

Air Boards

FTIR Gas Analyzer supplementary to their GC/MS for identifying and quantifying low molecular weight inorganic & organic gases

- **HCl**
- **HF,**
- **NH3**
- **HCHO**

- **Mounted in van or taken to field**

- **Advanced Library Search Function assisting them identify "Unknowns"**

Gasmet

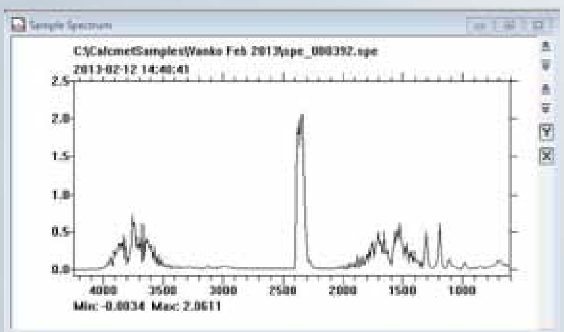
Case Study

Identify the Unknown

Freon Leak goes undetected

Analysis Results - GAS-LIB-004_SN112203_c11_11_Rev1.LIB: spe_000392.spe

Ch	Component	Concentration	Unit	Range	Resid...
1	Water vapor	0.62	vol-%	3	0.0004
2	Carbon dioxide	674.13	ppm	2000	0.0014
3	Carbon Monoxide	0.00	ppm	200	0.0015
4	Nitrous Oxide	0.30	ppm	100	0.0008
5	Sulfur Dioxide	0.00	ppm	100	0.4425
6	Nitrogen Dioxide	2.06	ppm	50	0.0022
7	Methane	1.84	ppm	100	0.0103
8	Formaldehyde	0.25	ppm	50	0.0010
9	Toluene	0.00	ppm	200	0.0101
10	Ammonia	2.93	ppm	50	0.0824
11	Hydrogen Chloride	0.00	ppm	50	0.0022
12	Hydrogen Fluoride	0.00	ppm	50	0.0006
13	Hydrogen Cyanide	0.00	ppm	50	0.0005
14	Arsine	0.01	ppm	50	0.0013
15	Phosgene	0.18	ppm	50	0.0020
16	Acrylonitrile	15.67	ppm	50	0.4521
17	Ethylene Oxide	0.15	ppm	50	0.0005
18	Boron Trichloride	0.00	ppm	50	0.4521
19	Phosphine	0.00	ppm	50	0.4671
20	Acrolein	0.00	ppm	50	0.0031
21	Methyl Mercaptan	32.35	ppm	200	0.0059
22	Carbon Disulfide	0.82	ppm	200	0.0009
23	Sulfuryl fluoride	0.00	ppm	50	0.4381
24	Dichloromethane	0.00	ppm	200	0.4495
25	Benzene	0.00	ppm	50	0.0144
226	Cell temperature	28.00	°C	40	0.0000



Calcmet Advanced Library Search

Library Search Results - 2013-02-12 14:40:41 C:\CalcmetSamples\Vanko Feb 2013\spe_000392.spe

Library	Component	Fit	Concentration
LibrarySearch_DX4030&40_c11_11_Rev4.LIB	Freon 134A	97.78	20.65
		0.00	0.00
		0.00	0.00
		0.00	0.00
		0.00	0.00
		0.00	0.00
		0.00	0.00
		0.00	0.00
		0.00	0.00

Healthcare facilities –
Complete solution for Toxic Gas

1. Operating & Recovery Rooms
WAG's – Desflurane, Nitrous Oxide, Sevoflurane, Enflurane, Isoflurane

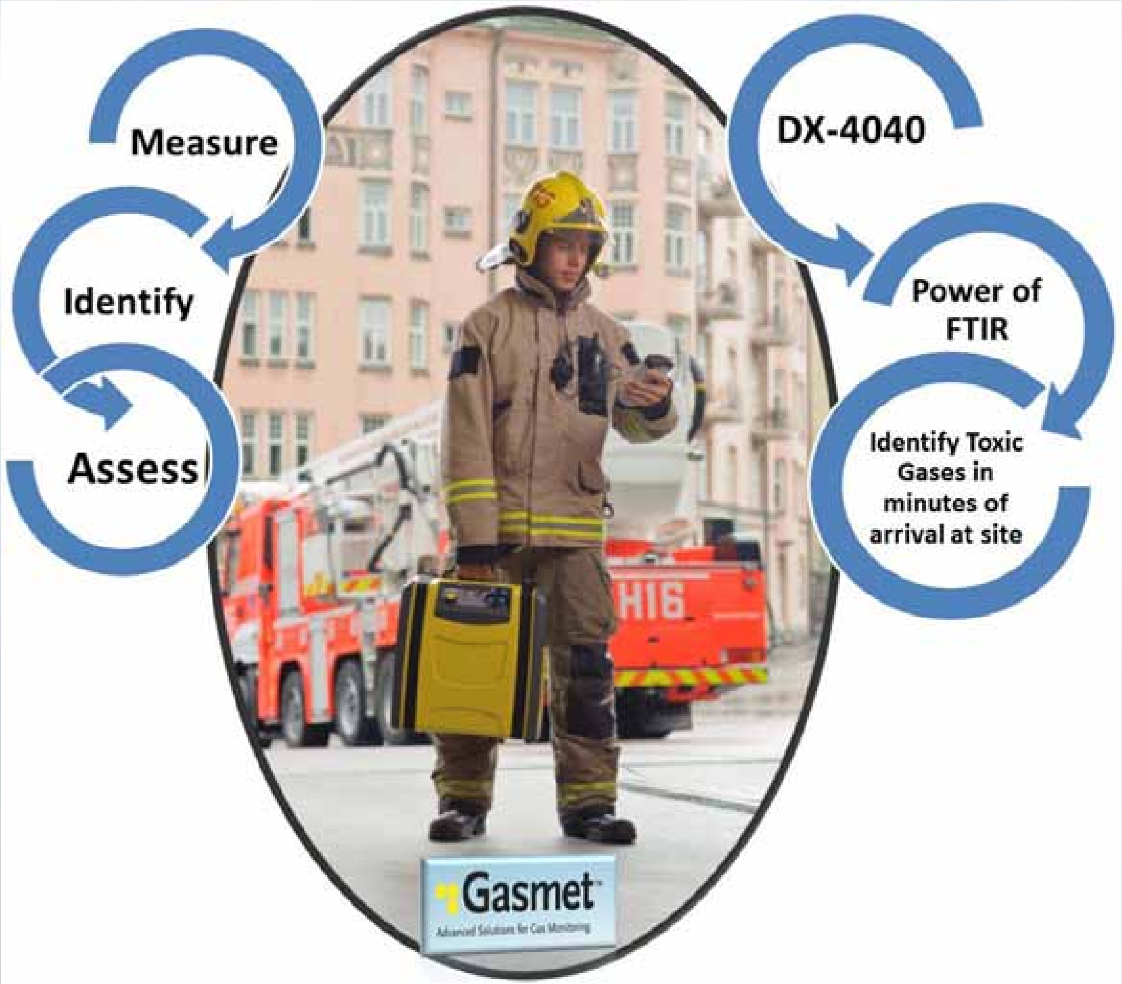
2. Toxic Sterilant Gases
a). Formaldehyde
b). Hydrogen Peroxide
c). Ethylene Oxide
d). Peracetic Acid

3. ED - Screening contaminated patients
(Identify the Unknown)

Climate Change Field Research

Government Research Scientists & University Researchers using Gasmet portable DX-series FTIR gas analyzer for the accurate measurement of GHG's in the field. Interdisciplinary research and publications using Gasmet FTIR gas analyzer have been published in Soil Science, Animal Husbandry, Landfill, Biomass and Agricultural Sciences from lead global research institutions.

Hazmat : Identifying the "Unknown"

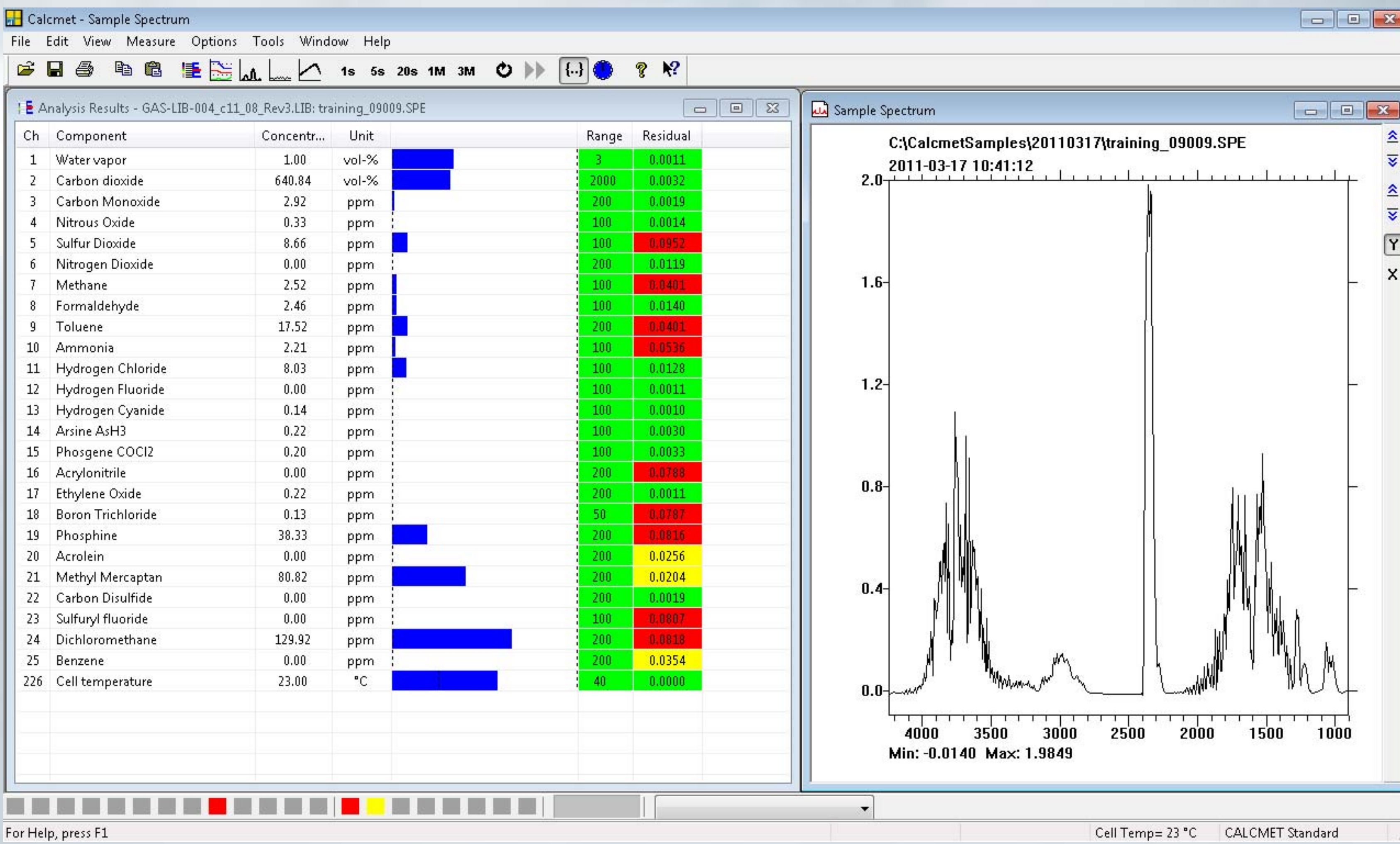


What if the responder has no information on the source of the gas or vapor ?

Time of Response is essential

Identifying “Unknown” Gases

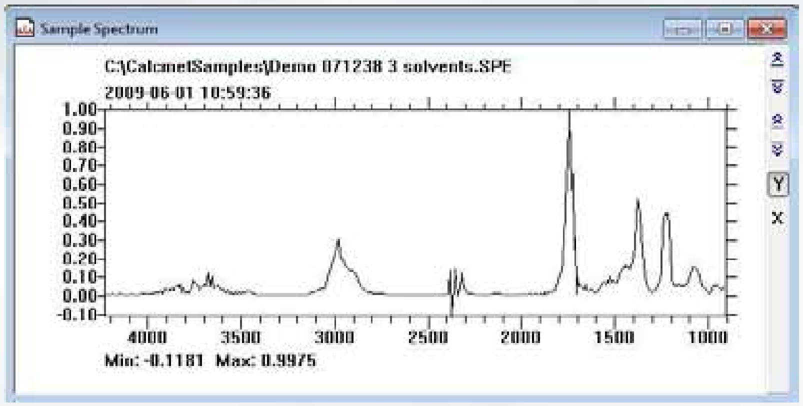
Step 1.



Ch	Component	Concentr...	Unit	Range	Residual
1	Water vapor	1.00	vol-%	3	0.0011
2	Carbon dioxide	640.84	vol-%	2000	0.0032
3	Carbon Monoxide	2.92	ppm	200	0.0019
4	Nitrous Oxide	0.33	ppm	100	0.0014
5	Sulfur Dioxide	8.66	ppm	100	0.0952
6	Nitrogen Dioxide	0.00	ppm	200	0.0119
7	Methane	2.52	ppm	100	0.0401
8	Formaldehyde	2.46	ppm	100	0.0140
9	Toluene	17.52	ppm	200	0.0401
10	Ammonia	2.21	ppm	100	0.0536
11	Hydrogen Chloride	8.03	ppm	100	0.0128
12	Hydrogen Fluoride	0.00	ppm	100	0.0011
13	Hydrogen Cyanide	0.14	ppm	100	0.0010
14	Arsine AsH3	0.22	ppm	100	0.0030
15	Phosgene COCl2	0.20	ppm	100	0.0033
16	Acrylonitrile	0.00	ppm	200	0.0788
17	Ethylene Oxide	0.22	ppm	200	0.0011
18	Boron Trichloride	0.13	ppm	50	0.0787
19	Phosphine	38.33	ppm	200	0.0816
20	Acrolein	0.00	ppm	200	0.0256
21	Methyl Mercaptan	80.82	ppm	200	0.0204
22	Carbon Disulfide	0.00	ppm	200	0.0019
23	Sulfuryl fluoride	0.00	ppm	100	0.0807
24	Dichloromethane	129.92	ppm	200	0.0818
25	Benzene	0.00	ppm	200	0.0354
226	Cell temperature	23.00	°C	40	0.0000

Identifying "Unknown" Gases

The Power & Speed of FTIR !



Advanced Library Search

Search Reference Library, over 350+ gases

Search NIST/EPA Library, over 5000 gases

Library Search Results - 2009-06-01 10:59:36 C:\CalcmetSamples\Demo 071238 3 solvents.SPE

Library	Component	Fit	Concentration
Librarysearch.LIB	Acetone C3H6O	97.46	144.58
Librarysearch.LIB	Ethanol C2H5OH	96.09	41.52
Librarysearch.LIB	Isopropanol C3H8O	93.86	26.02

Operating a FTIR gas analyzer

Step 1: FTIR Gas Analyzer : Operation

Step 1) AC Power or Battery power

Step 2). Warm-up – no time if plugged into AC outlet
~5 mins if instrument "cold"

Step 3). Background or zero calibration.

- Connect Nitrogen gas (as illustrated)
- Operator pushes ZERO measurement on PDA
- Automatically Zeroing DX4040 (takes 3 mins)

No Span gas or other consumables required

Instrument ready for full day of testing

Step 2. Operation – Taking measurements

Gas measurements - updated each 5,20 or 60 seconds
- Individual audible and visual alarm (user adjustable)
- Results as text file, download via USB port, open directly into Excel® spreadsheet, Tag file name.
- Up to 25 gases measured simultaneously
- WiFi or 3G data to central command
- Includes GPS coordinates, photos or video

Validation & Certification

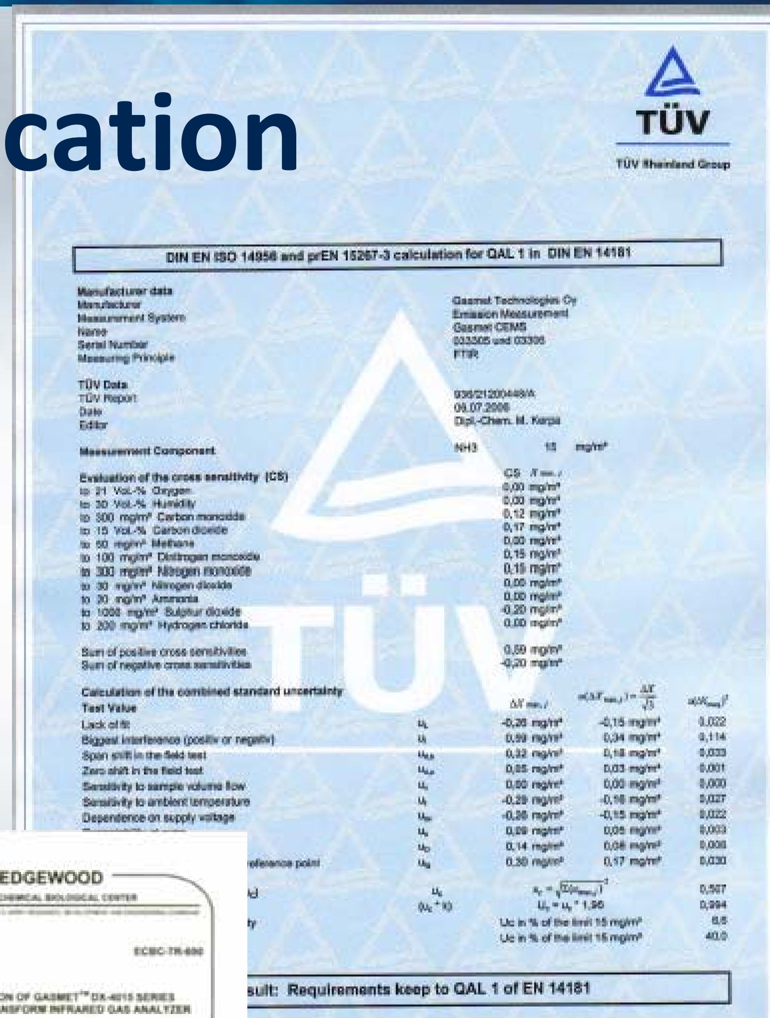
FTIR Methodology accepted by leading test organizations including :

- **NIOSH** Method 3800 (Organic & Inorganic gases by extractive FTIR spectrometry)
- **USEPA** Method 320
- **ASTM** Method D6348

- TUV & MCERTS 3rd party verification
- US Military Edgewood Report

Measurement accuracy

- Linearity < 2% range
- Sum cross interferences < 4% of the range


TÜV
TÜV Rheinland Group
DIN EN ISO 14956 and prEN 15267-3 calculation for QAL 1 in DIN EN 14181

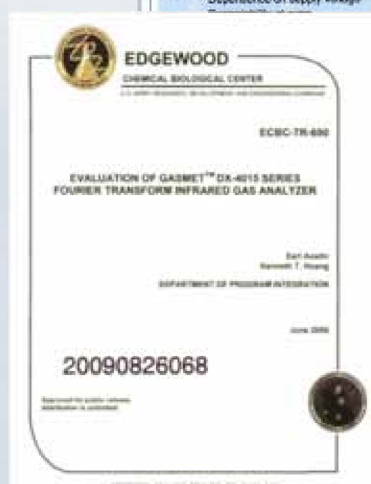
Manufacturer data	
Manufacturer	Gasmet Technologies Oy
Measurement System	Emission Measurement
Name	Gasmet CEMS
Serial Number	033305 und 03306
Measuring Principle	FTIR
TÜV Data	
TÜV Report	936/21200448/A
Date	08.07.2006
Editor	Dipl.-Chem. M. Kerpa
Measurement Component	NH3 15 mg/m³

Evaluation of the cross sensitivity (CS)

Sum of positive cross sensitivities 0,59 mg/m³
Sum of negative cross sensitivities -0,20 mg/m³

Calculation of the combined standard uncertainty

Result: Requirements keep to QAL 1 of EN 14181


EDGEWOOD
CHEMICAL BIOLOGICAL CENTER
ECBC-TR-690
EVALUATION OF GASMET™ DX-4015 SERIES FOURIER TRANSFORM INFRARED GAS ANALYZER
DEPARTMENT OF PROGRAM INTEGRATION
20090826068

Instrument Maintenance

Low Cost
of
Ownership

<u>**Zero gas (Nitrogen, purity 5.0 (99.999%)**</u>

Capacity : 552L.   Typical usage per zero : 10L
~ 1 year / cylinder based on weekly usage.

Other gas cylinder sizes available. Easy local supply

<u>**Particulate Filters**</u>

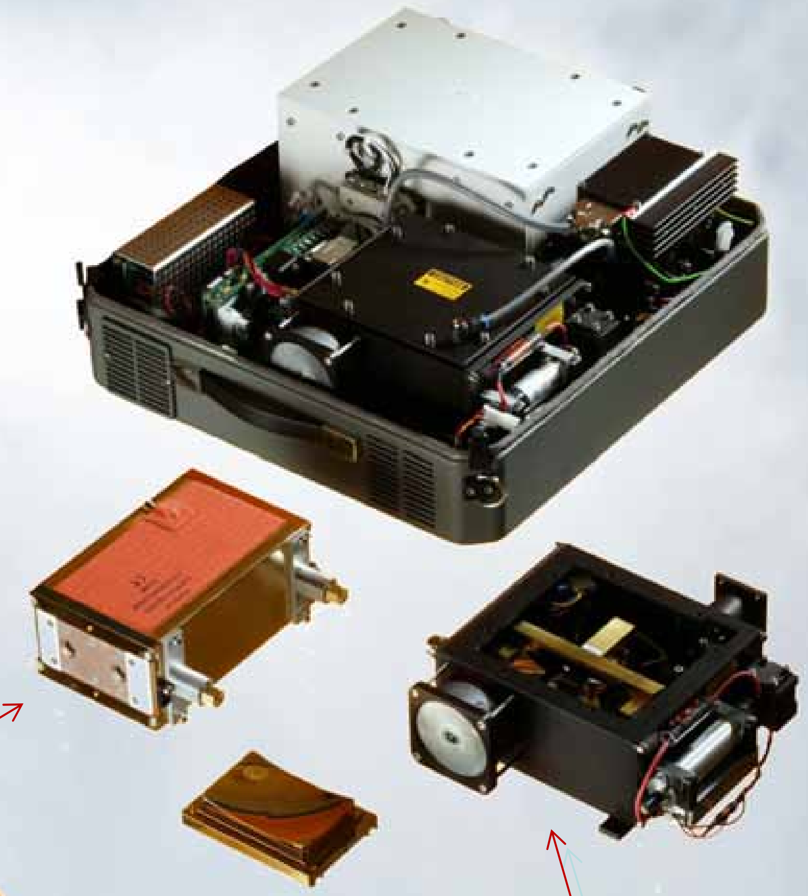
Weekly visual inspection of sample probe filter (change the filter, if necessary). Only dusty applications require continual filter change.

Anticipate filter change out for each 3 months.
Set of 5 spare filters supplied

No requirement to send instrument back to manufacturer for annual service

Gasmet

Introducing the FTIR Gas Analyzer

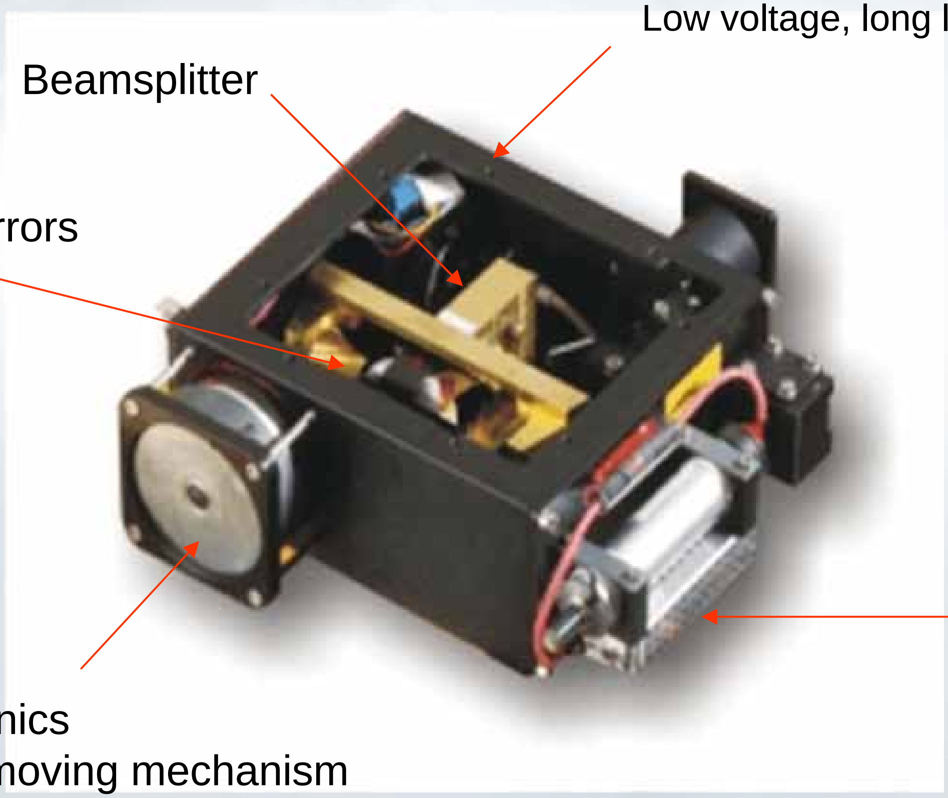


Sample Cell

Interferometer

Giccor interferometer

Compact size – Rugged – Fast scanning – Excellent stability

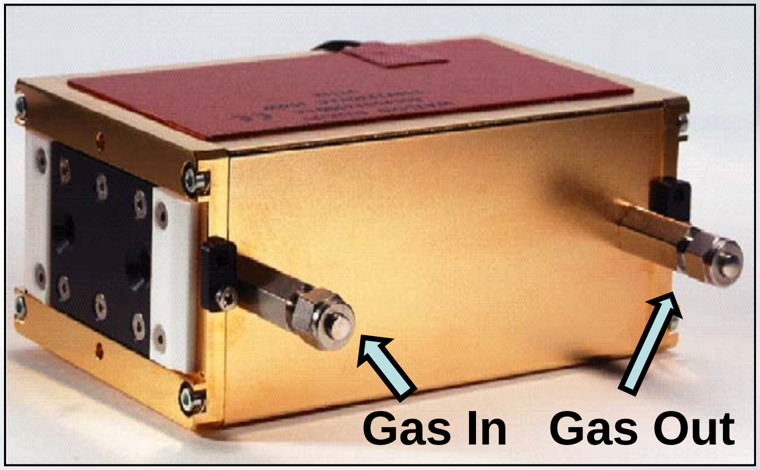


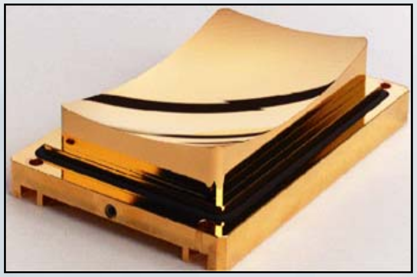
Low resolution (8 cm^{-1}) means better signal to noise ratio as optical aperture is larger (more light gets through)

Corrosion resistant sample cells

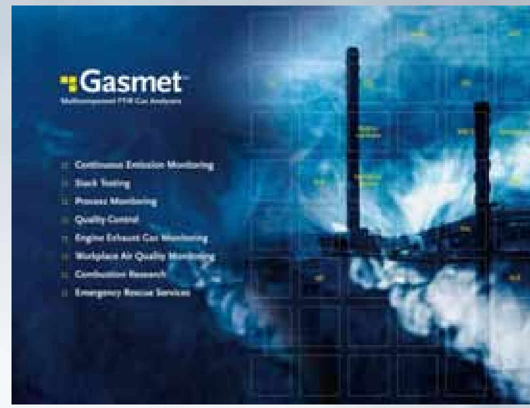
- **Inert -** Nickel-rhodium-gold plated mirrors & aluminium cell body
- **Fixed mirrors** (resilient to shocks & bumps of handling & transporting)
- Same **rugged tough gas cell** used in CEMS as ambient application
- Absorption to 9.8 m Single pass and multi-pass (White cell)
- Cell windows (ZnSe, immune to water vapour)
- **Gas Measurement Range** :

  Sub-ppm to % levels





Experience in FTIR Gas analysis



- Established in 1990 in Helsinki, Finland
- Manufacturing FTIR gas analyzers, their accessories and systems
- Distributors & Representatives in 67 countries
- > 2500 analyzers or systems delivered to date
- Brought to market innovative products :
- First In-Situ FTIR in 2003 & First truly portable FTIR (DX4030) in 2008

- US customers include:

• US DoD	• Lockheed Martin	• Dow Corning
• NASA – Kennedy Space Center	• Suncor Energy	• Marshfield Clinics
• **NIOSH**	• MI Region 2	• Dow Chemicals
• US DoE - Hanford Site	• Caterpillar Corporation	• DuPont
• US Dept. of Agriculture	• Utah State University	• Texas A & M
• University of Cincinnati	• Hemlock Semiconductor	• Starbucks

Benefits of applying FTIR in Routine & Hazmat multi-gas testing applications

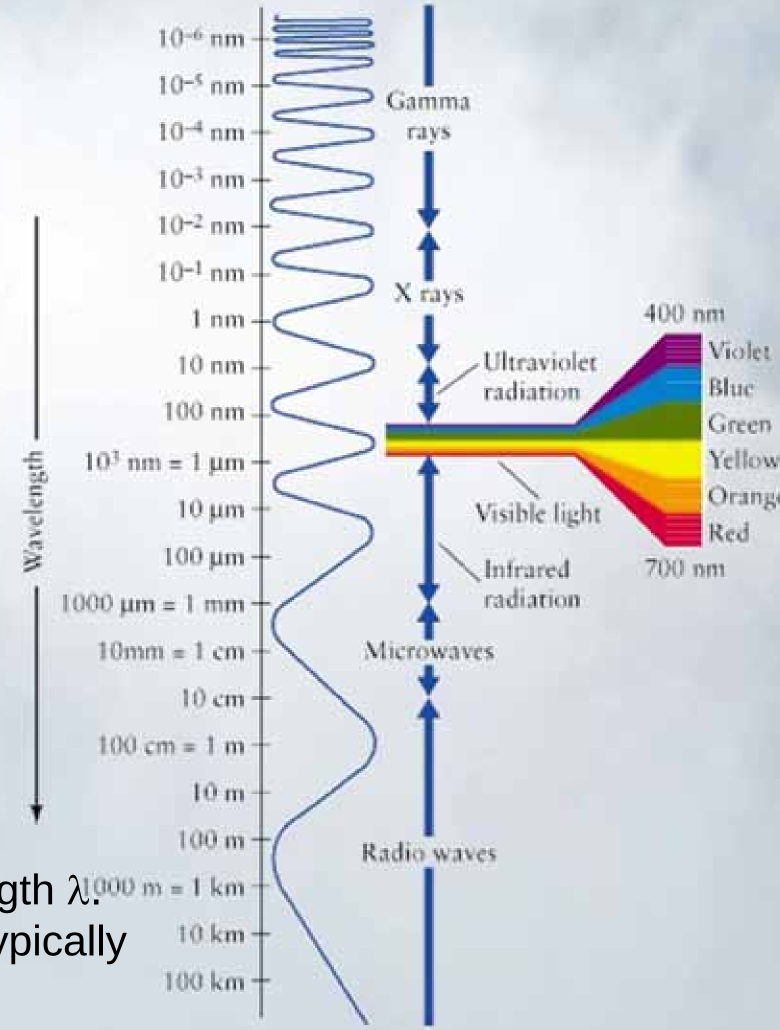
- **Fast response** ***& direct reading to 200+ chemicals*** – ***25 gases measured simultaneously*** available to the user in a just minutes from start-up.
- **Ease of Use** - Allows multiple users to be trained resulting in effective usage of the equipment
- **Validation of results** – Laboratory Accuracy in the field - FTIR NIOSH approved method, all results (tamperproof) .
- **Low Cost of Ownership** [No water removal, no sample preparation, no carrier gases, no calibration gases (internal calibration using laser)]
- **Identify "Unknown" Gases –** Software quickly searches over 5,000 gases for positive id of toxic gases

Introduction to FTIR Theory

Electromagnetic radiation

Radiation type	**Wavelenght, λ**
Radio waves	> 30 cm
Micro waves	1 mm - 30 cm
Infrared Mid-IR (used by Gasmet)	700 nm - 1 mm 2.5 µm – 10 µm (4000-1000 cm^{-1})
Visible light	350 nm - 700 nm
UV light	10 nm - 350 nm
X-ray	0.01 nm - 10 nm
Gamma rays	< 0.01 nm

$$f = \frac{c}{\lambda} \qquad \tilde{\nu} = \frac{f}{c} = \frac{1}{\lambda}$$

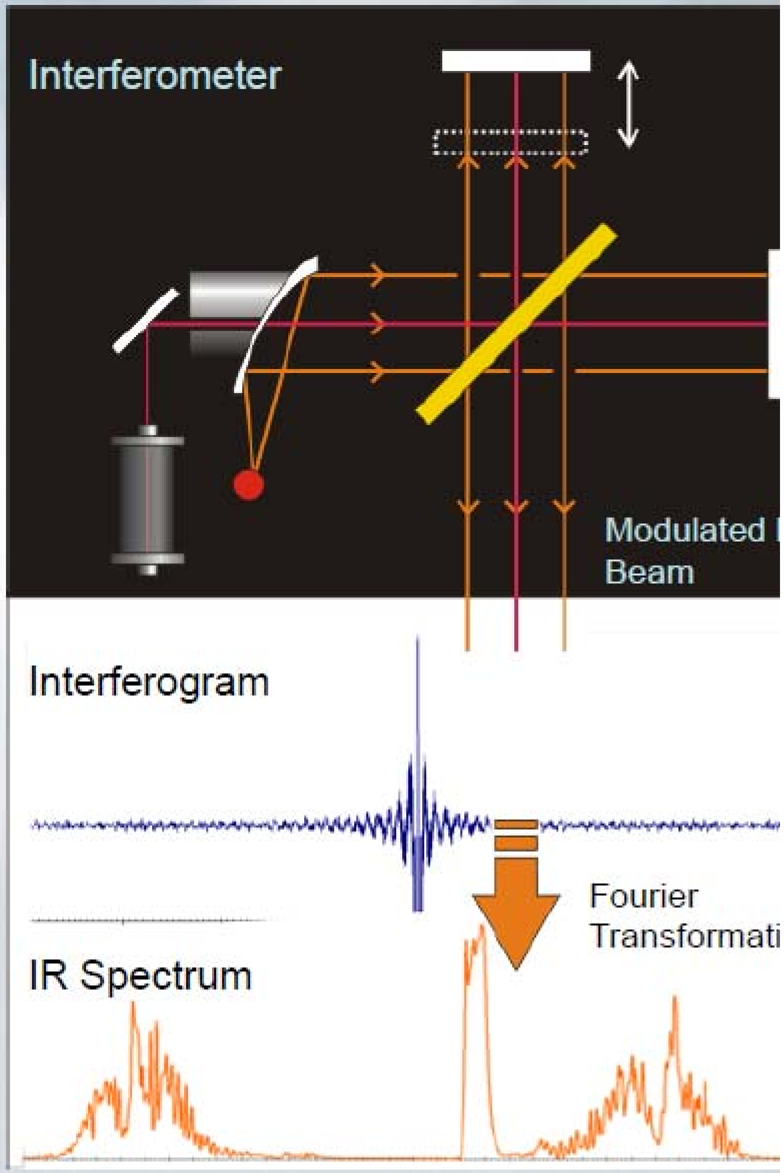


Frequency (f) is speed of light (c) divided by wavelength λ.
Wavenumber $\tilde{\nu}$ is the inverse of wavelength and is typically expressed in the units of reciprocal centimeter (cm^{-1})

Gasmet

FTIR Spectroscopy

- Based on the use of an optical modulator: *interferometer (Michelson, circa 1890's)*
- Interferometer modulates radiation emitted by an IR-source, producing an *interferogram* that has all infrared frequencies encoded into it
- Interferometer performs an optical Fourier Transform on the IR-radiation emitted by the IR-source (Jean Fourier 1768-1830)
- The whole infrared spectrum is measured at high speed (10Hz)
- Spectral range is continuously calibrated with He-Ne laser [Internal calibration]
- Fast, extremely accurate measurements

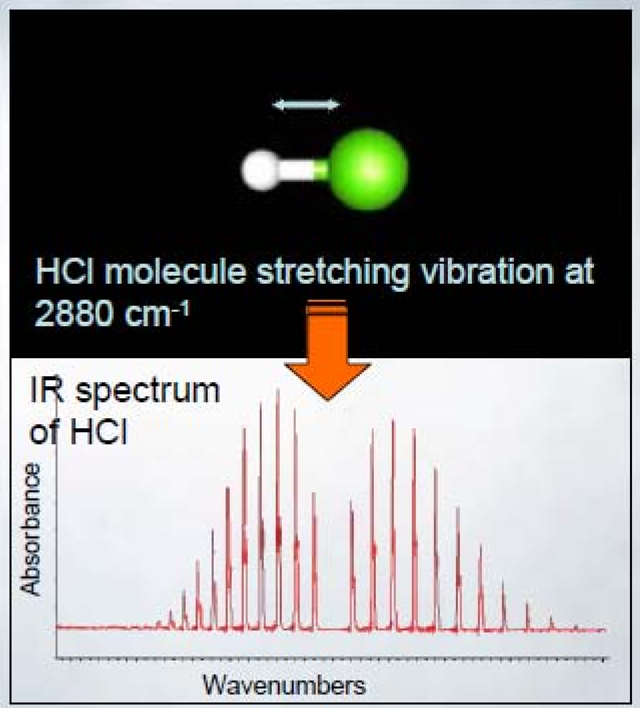


Gas phase infrared spectroscopy

- Molecules in gas phase vibrate and rotate at frequencies characteristic to each molecule. Each frequency is associated with an energy state of a molecule
- Infrared radiation moves the molecules to higher energy states; characteristic frequencies are absorbed by the molecule in the process
- Each molecule absorbs infrared radiation at several characteristic frequencies (wavelengths)
- The result is an IR absorption spectrum; a fingerprint unique to each molecule

Gas phase infrared spectroscopy

- All molecules can be identified on the basis of their characteristic absorption spectrum (except diatomic elements such as O_2 and noble gases)
- Each molecule absorbs infrared radiation at its characteristic frequencies
- IR absorption spectrum is a fingerprint unique to each molecule
- Beer's law: Absorption strength i.e absorbance is directly proportional to concentration

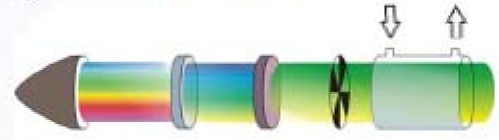


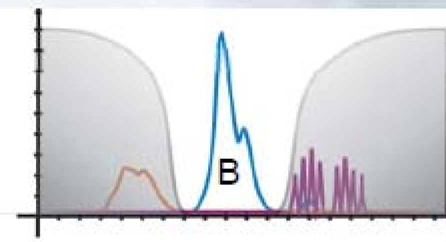
All gases except O_2, N_2, H_2, Cl_2, F_2, H_2S, and noble gases can be measured

IR technologies

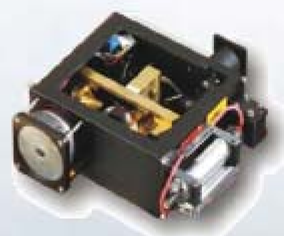
- **Gas Filter Correlation IR (GFC)**
  - Measures only separate wavelength bands with gas-filled filters
  - Only one component can be measured with each filter
  - Multiple gases can be measured and spectral interference resolved only with additional filters (typically maximum 6 gases)
  - Multiple gas filled filters means multiple calibration checks

Broad band light source | Optical filters | Sample cell

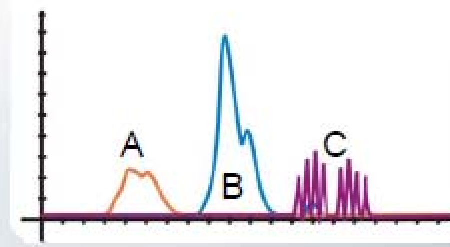




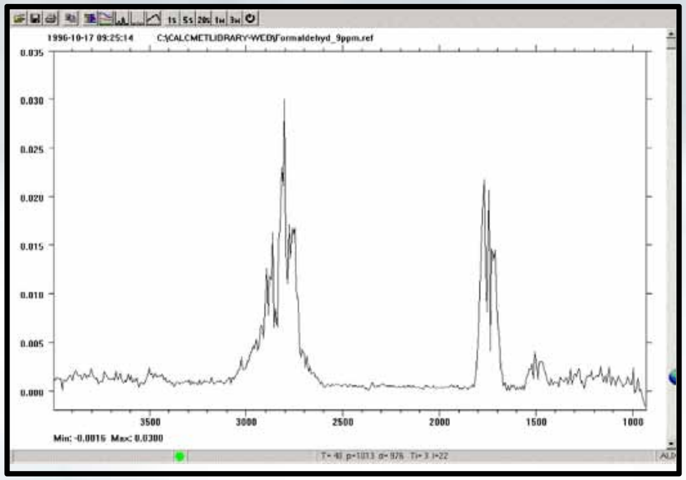
- **Fourier Transform Infrared (FTIR)**
  - Spectrometer measures all the IR wavelengths simultaneously and produces a full spectrum.
  - Any number of components (up to 50) can be analysed from single measurement and interferences are automatically resolved
  - Same optical elements used for each measurement, multiple calibration checks are not necessary



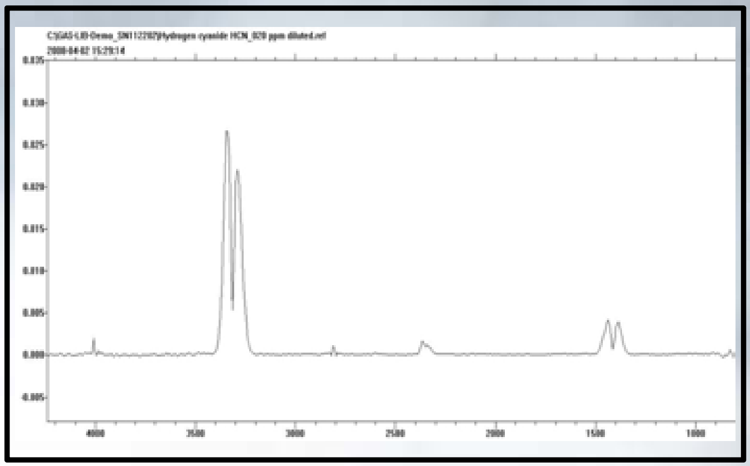
Interferometer



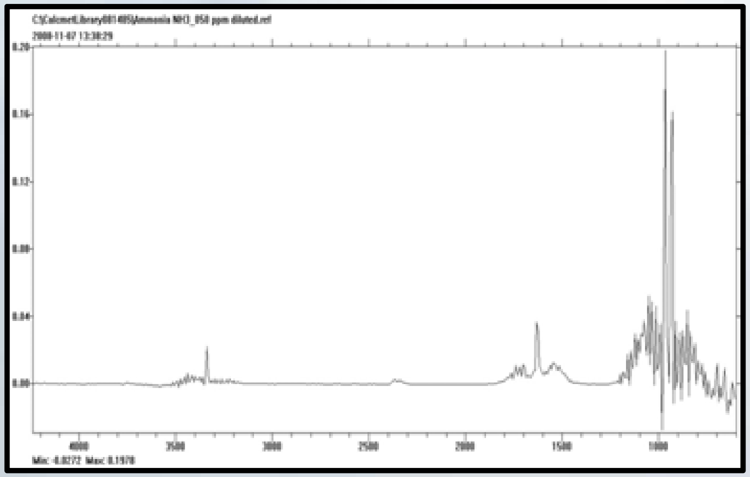
Examples of Infrared Spectra



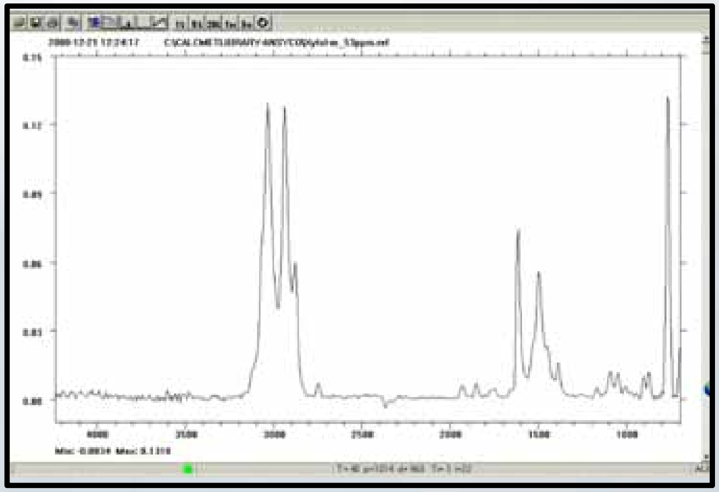
Formaldehyde - HCHO



Hydrogen Cyanide– HCN



Ammonia – NH_3



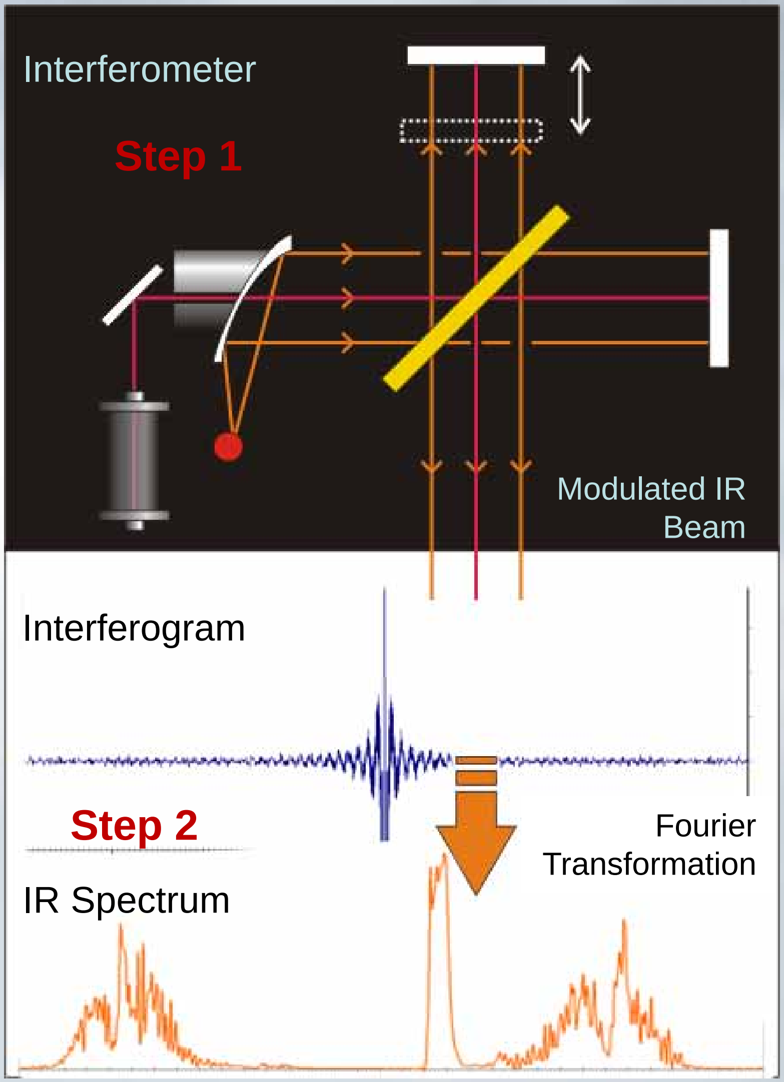
Benzene – C_6H_6

FT-IR Spectroscopy

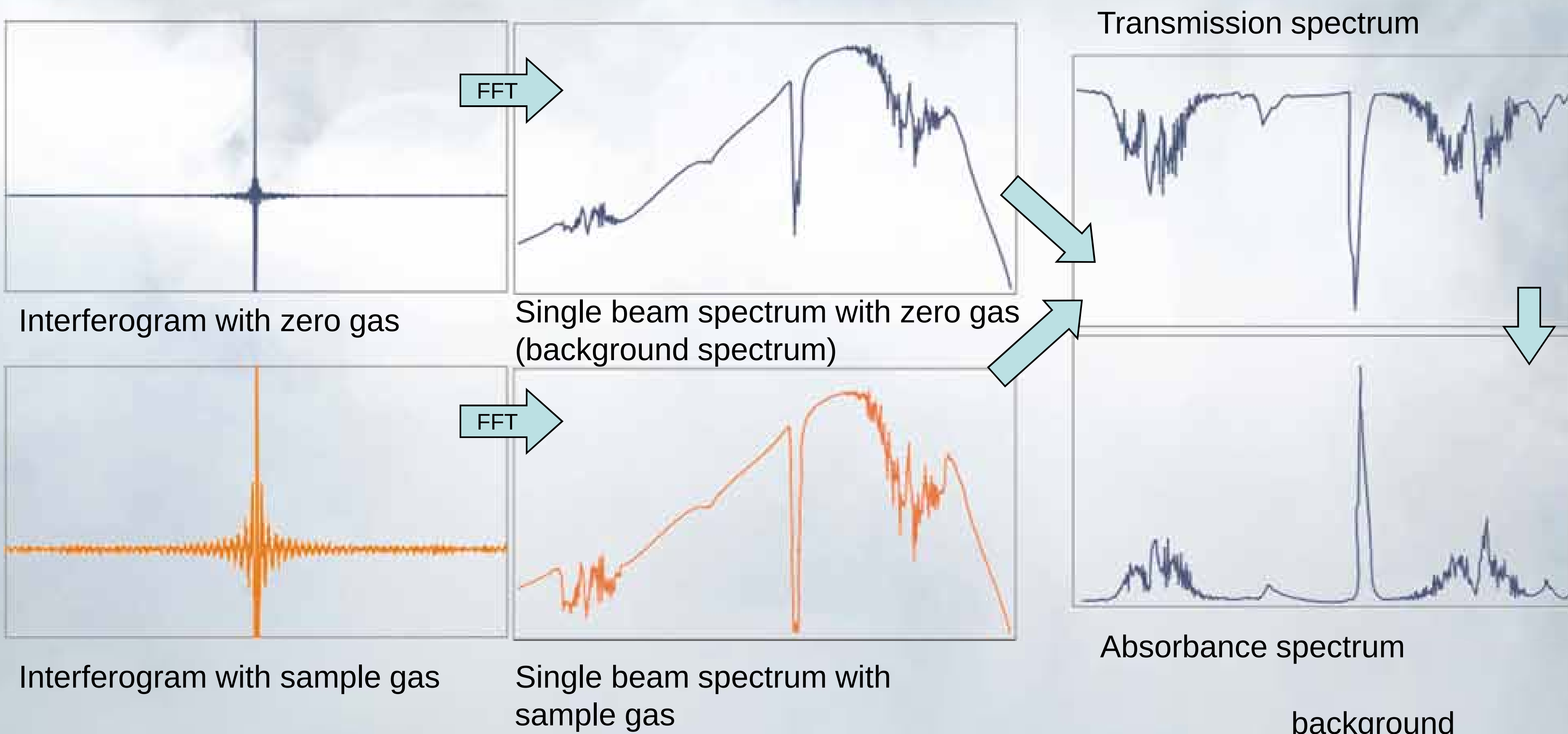
- Based on the use of an optical modulator: *interferometer (Michelson, circa 1890's)*

Two Steps –

- **1).** Interferometer modulates radiation emitted by an IR-source, producing an *interferogram* that has all infrared frequencies encoded into it
- **2).** Apply mathematics (Fourier Transform) to produce IR Spectrum
- IR spectrum measured 10 cycles/sec (10Hz)
- **Internal calibration** with precise laser
- <u>Daily Zero calibration, no recalibration required</u>
- Fast, extremely accurate measurements

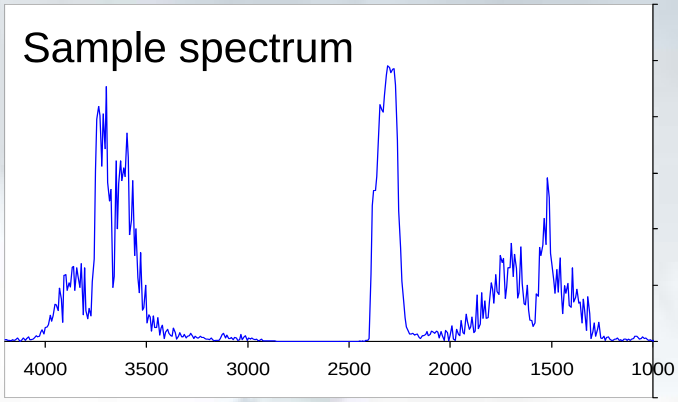


Measuring Sequence

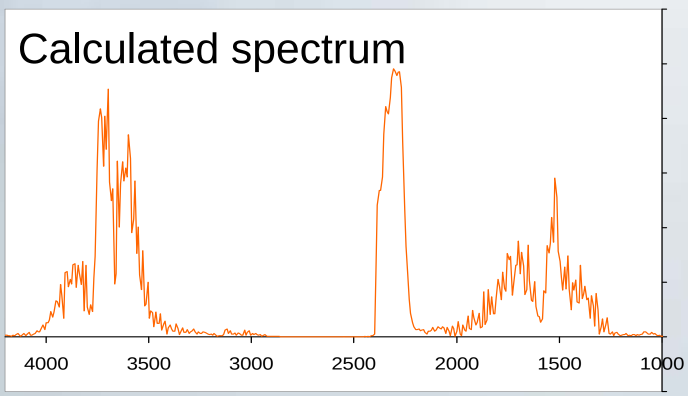


$$A = -\log_{10}\frac{\text{background}}{\text{sample}}$$

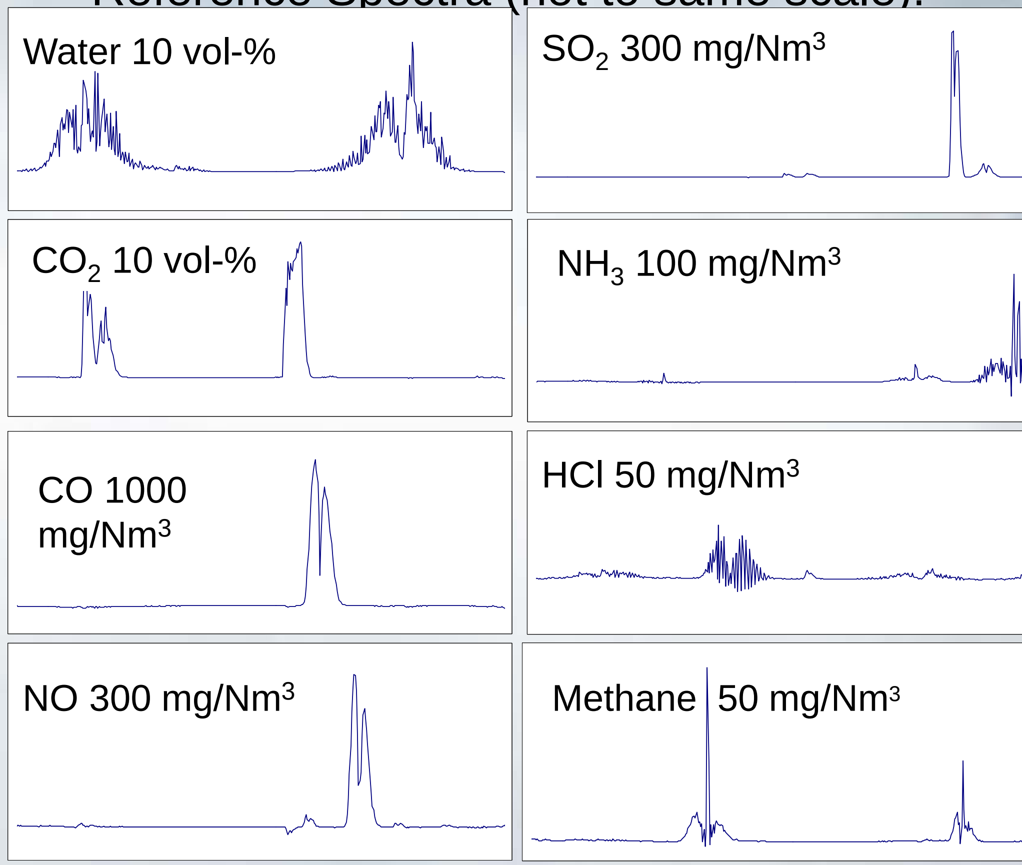
CALCMET analysis:



CALCMET Analysis:
0.881 * Water 10 vol-%
1.112 * CO_2 10 vol-%
0.995 * CO 1000 mg/Nm^3
0.910 * NO 300 mg/Nm3
0.810 * SO_2 300 mg/Nm3
0.660 * NH_3 100 mg/Nm^3
0.082 * HCl 50 mg/Nm^3
0.210 * Methane 50 mg/Nm^3



Reference Spectra (not to same scale):



Concentrations:			
Water	8.81 vol-%	SO_2	243 mg/Nm^3
CO_2	11.12 vol-%	NH_3	66.0 mg/Nm^3
CO	955 mg/Nm^3	HCl	4.1 mg/Nm^3
NO	274 mg/Nm^3	Methane	10.5 mg/Nm^3

Gasmet

Simultaneous Gas Readings

Analysis Results - GAS-LIB-007_c11_101_Rev7.LIB: OR7 Desflurane_00021.spe

Measured Components

Concentration

Bar graph display

Calibration Range

Residual value

Audio & Visual Alarms / gas

Up to 25 gases →

Ch	Component	Concentration	Unit	Range	Residual
1	Water vapor	0.60	vol-%	3	0.0013
2	Carbon dioxide	411.58	ppm	1000	0.0024
3	Carbon monoxide	0.11	ppm	100	0.0004
4	Methane	1.76	ppm	50	0.0013
5	Nitrous oxide	1.62	ppm	50	0.0011
6	Sevoflurane	0.02	ppm	50	0.0013
7	Desflurane	3.47	ppm	20	0.0010
8	Isoflurane	0.03	ppm	50	0.0010
9	Halothane	0.04	ppm	50	0.0010
10	Enflurane	0.02	ppm	50	0.0009
11	Ethylene oxide	0.00	ppm	10	0.0008
12	Formaldehyde	0.03	ppm	50	0.0007
13	Glutaraldehyde	0.00	ppm	20	0.0007
15	o-Phthaldehyde (OPA)	0.00	ppm	10	0.0008
16	Methylmethacrylate	0.02	ppm	50	0.0012
17	Ammonia	0.09	ppm	100	0.0007
18	Methanol	0.13	ppm	100	0.0012
19	Ethanol	0.52	ppm	200	0.0009
20	Isopropanol	0.11	ppm	200	0.0013
21	Acetone	0.00	ppm	200	0.0017
22	Toluene	0.47	ppm	100	0.0012
23	m-Xylene	0.00	ppm	200	0.0018
24	o-Xylene	0.05	ppm	200	0.0013
25	p-Xylene	0.00	ppm	200	0.0017

Thank you !!

Joel Myerson

Joel@ESafetyInc.com

978-532-7330